KEN SHIBATA

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Education

Georgia Institute of Technology May 2028

- Bachelor of Science in Computer Science (Theory and Systems & Architecture) (4.00 GPA)
- Coursework: Algorithms, Discrete Math, Multivariable Calculus, Parallel Computing Applications

Work Experience

Software/Firmware Intern, Marvell Semiconductor Canada Inc. *Toronto, ON* May - Aug 2025

- Created a multicore and generic debugger to support 8051 development user for high-speed SerDes IP
- Improved a unit test and introspection framework based on high-layer emulation of an embedded MCU

Undergraduate Research Assistant, Georgia Tech (GT) Open Source Program Office Jan 2025 - present

- Writing documentation to help researchers easily preserve HPC programs for long-term reproducibility
- Developing software to run virtual machines on-demand (without waiting for the VM to download)

Web Development Intern, Helpfeel, Inc. *Tokyo, Japan* July 2024

- Worked on Helpfeel Cosense, a Node.js and React-based knowledge base wiki editor with 410k users
- Implemented UX improvements and English text input for Android and iOS

Selected Projects

VR State Analysis, Georgia Tech Extended Reality club and Big Data club Nov 2024 - present

- Leading a team of ten to detect stress factors using VR headsets, for integration in VR games
- Heart rate variability detection using ML and only VR headset IMU/motion data (no extra hardware)

HATO: Immersive Automatic Model Train Control System Apr - Nov 2023

- Hardware/software to fully automate 1:150 scale model trains using Go and C
- Presented to 200+ at the University of Tokyo; awarded the Supercreator (excellence) distinction

Yarnkey: Wearable Keyboard using Conductive Threads-based Touch Mar 2021 - Aug 2022

- Created Yarnkey to use in crowded buses; received funding for materials and travel
- Supported by NII Experts in Information Science program; Presented at DICOMO 2022 and WCCE 2022

Leadership / Community Engagement / Awards

Full-ride Scholarship

- Awarded full-ride 4-year scholarship for undergraduate studies from the Masason Foundation

Coding Lead, Cognitive Load Project, Brain Trauma Assessment Protocols VIP team Jan 2025 - present

- Developing video-based ML models to detect and diagnose aphasia patients' cognitive load (e.g. rPPG)
- Working under Prof. Elliot Moore, with a team of twenty CS and Biomedical Engineering students
- Oral presentation at the 19th Georgia Tech Undergraduate Research Spring Symposium

Ground Station Developer, Ramblin' Rocket Club Aug 2024 - present

- Integrating and developing software/hardware for a mission control station using C and Python
- Oral presentation at American Institute of Aeronautics and Astronautics Region 2 Student Conference

Skills and Interests

- Programming Languages (fluent): Go, C, Python, Java, JavaScript, OCaml, Nix
- Software: GTK 4, SQLite, Django, Postgres, React.js, Svelte, Git, WireGuard, systemd
- Hardware: 3D printing, CAD, microcontrollers (STM32, Arduino), embedded (MCS-51)
- Languages (native): English and Japanese
- Hobbies: model railroading (3D printing), photography, free and open source software

Last updated Aug 2025.